

Cube Roots — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 • MCQ**[1]**

$$\sqrt[3]{1000} = ?$$

- (a) 10 (b) 100 (c) 1 (d) none of these

Q2 • MCQ**[1]**

$$\sqrt[3]{729} = ?$$

- (a) 9 (b) 8 (c) 27 (d) 3

Q3 • FILL IN THE BLANK**[1]**

$$\sqrt[3]{64} =$$

Q4 • MCQ**[1]**

$$\sqrt[3]{512} = ?$$

- (a) 6 (b) 7 (c) 8 (d) 9

Q5 • TRUE / FALSE**[1]**

$$\sqrt[3]{27} = 3.$$

True False

Q6 · ASSERTION-REASON**[1]**

Assertion (A): The ones digit of the cube root of 1728 is 6.

Reason (R): The cube root of a number is the factor that is multiplied by itself three times to give the number.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)**Q7****[2]**

Find $\sqrt[3]{2197}$ by prime factorisation.

Q8**[2]**

Find $\sqrt[3]{216}$ by prime factorisation.

Q9**[2]**

Find $\sqrt[3]{5832}$ by prime factorisation.

Long answer (3 marks each)**Q10****[3]**

Find $\sqrt[3]{1728}$ by prime factorisation, showing the grouping in threes.

Q11

[3]

Find $\sqrt[3]{13824}$ by prime factorisation.

Q12

[3]

A cubical box has a volume of 3375 cubic centimetres. Find the length of its edge, showing your method.

Total: 21 marks · Check your work against the [answer key](#).